Aum Amriteshwaryai Namah

25CSA511A Data Structures and Algorithms 3 0 2 - 4

Course Objectives

This course aims to provide basic knowledge of different data structures and its usage. It also covers techniques used for analyzing algorithms and notations for expressing time complexity.

Course Outcomes

COs	Description
CO1	Implement basic data structures such as Linked lists, Stack and Queue.
CO2	Analyze the time and space complexities of algorithms to evaluate their efficiency in various applications.
CO3	Implement different searching and sorting algorithms.
CO4	Use different data structures including tree and graph and solve computational problems using it.
CO5	Evaluate different programming paradigms to choose appropriate techniques for problem-solving.

CO-PO Mapping

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	-	1	-	-	-	2
CO2	3	1	-	1	-	-	-	3
CO3	3	3	-	1	-	-	-	1
CO4	3	3	-	1	-	-	-	3
CO5	2	3	-	2	-	-	-	3

3-strong, 2-moderate, 1-weak

Syllabus

Unit I

Linear Data Structures: Arrays (single and multi-dimensional), Stack ADT, Multi Stack ADT, Queue ADT, Circular Queue, Priority Queue, Singly Linked List, Doubly Linked List, Circular Linked List.

Unit II

Nonlinear Data Structures: Trees - Array and List Representations: Binary Tree, Binary Search Tree, and Threaded Binary Tree. Balanced Trees: Weight Balanced Trees, Applications of WBTs, Height Balanced Trees -AVL Trees, Red-Black Trees. Binary Heaps: applications

Unit III

Graphs: Matrix and List Representation of Graphs, Breadth-First Search, Depth First Search, Shortest Path Algorithms (with Analysis) – Dijkstra - Bellman Ford- Floyd Warshall's all Pair shortest path Algorithm-Minimum spanning Tree (with Analysis) – Kruskal– Prims - Applications of BFS and DFS.

Unit IV

The efficiency of algorithms - the notion of time and space complexity, Basic Complexity Analysis - Worst case, Average case, and Best cases, Asymptotic Analysis- notations, analyzing iterative programs – Simple examples; Recurrences, Recurrence Relation: Substitution method, Recursion Tree Methods, Master Method.

Unit V

Dynamic Programming: Longest common subsequence problem, Matrix Multiplication Problem- 0/1 Knap-sack Problem. Branch and Bound – backtracking , Analysis of Divide and conquer algorithms – Quick Sort, Merge sort, Bucket Sort, Heap Sort, Greedy Algorithm: Fractional Knapsack Problem- Task Scheduling Problem.

Textbooks / References

1. Ellis Horowitz, Sartaj Sahni, and Susan Anderson-Freed, "Fundamentals of Data Structures in C", Second Edition, Silicon Press, 2008.

2. Jean-Paul Tremblay and G. Sorenson, "An introduction to Data Structures with Applications", Second Edition, Tata McGraw-Hill, 2008.

3. Robert L.Kruse, Bruce P. Leung, Clovis.L. Tondo and Shashi Mogalla, "Data Structure and Program Design in C", Pearson Education, Second Edition, 1997.

4. CormenT.H, Leiserson C.E, Rivest R.L, and Stein C, "Introduction to Algorithms", Third Edition, Prentice-Hall of India, 2009.

5. Baase.S and Gelder A.V., "Computer Algorithms- Introduction to Design and Analysis", Third Edition, Pearson Education Asia, 2003.

AMRITA VISHWA VIDYAPEETHAM Amrita Online - Course Plan					
Week	**Topic**	**Learning Objectives**	**eLearning Content**	**Video Lectures**	**Assessments**
1	**Linear Data Structures**	Introduction to data structures, different types	6 Videos 1 Quiz	1) Introduction to data structures. 2) Arrays – 1D 3) Multi-dimensional Arrays 4) Stack ADT 5) Implementing two stacks 6) Implementing Multistack	Quiz 1
2	**Applications of Stack And Queue**	Understand Stack applications and queue	6 Videos 1 Quiz	1) Stack Applications - Function call, Recursion 2) Postfix evaluation 3) Infix to postfix 4) Queue 5) Circular Queue 6) Priority Queue	Quiz 2
3	**Linked List**	Understand Singly Linked list, and its operations.	3 Videos 1 Lab sheet 1 Quiz	1) Linked list 2) Linked list implementation and operations 3) Linked list loop detection	Quiz 3 Theory Asgmt 1
4	**Types of LL and Trees**	Understand Types of Linked List and tree	5 videos 1 quiz 1 asgmt	1) Circular Linked List 2) Doubly Linked List 3) Tree- tree applications, binary tree 4) Array representation of Tress 5) Linked List representation of tress, Binary tree insertion and traversal.	Quiz 4
5	**Trees**	Familiarize Binary search tree	4 videos 1 quiz 1 lab sheet	1) Binary search tree 2) Binary Search tree – search, insertion 3) BST -Deletion 4) AVL tree	Quiz 5 Lab Asgmt 1

6	**Graphs**	Understand Graphs	4 Videos 1 quiz 1 labsheet	1) Introduction to Graphs 2) Representation of graph data structures. 3) Graph traversal -BFS 4) Graph traversal,DFS	Quiz 6

7	**Graphs**	Understand Minimum spanning and Shortest path	3 videos 1 quiz 1 assgmt	1) Kruskal Algorithm 2) Prim's Algorithm 3) Dijkstra's algorithm	Quiz 7 Theor y Asgm t 2
8	**Algorithm Analysis**	Understand Introduction to algorithms	4 videos 1 quiz Labsheet	1) Bellmanford 2) Floydwarshalls 3) Introduction to algorithm analysis 4) Analyzing algorithms-Best, worst, average case analysis	Quiz 8
9	**Algorithm Analysis**	Understand Analysis of Iterative and recursive programs.	3 videos 1 quiz 1 lab sheet	1) Asymptotic noatations 2) Analysis of Iterative programs. 3) Analysis of Recursive program.	Quiz 9
10	**Divide and conquer**	Familiarize Quick sort analysis	1 videos 1 quiz	1) Solving Recurrence using masters method	Quiz 10
11	**Algorithm design and quick sort**	Familiarize Quick sort	3 videos 1 asgmt	1) Algorithm design techniques 2) Quick sort 3) Quick sort analysis	Quiz 11 Lab Asgmt 2

12	**Quick and Merge sort**	Familiarize Merge sort and quick sort	2 videos	1) Merge sort 2) Quick sort demonstration.	Quiz 12
13	**Dynamic Programming**	Understand Greedy and dynamic programming.	2 videos 1 quiz	1) Greedy Approach+ Knapsack problem 2) Dynamic Programming -introduction.	Quiz 13
14	**Dynamic Programming**	Understand Knapsack problem using Dp.	3 videos 1 quiz	1) Knapsack problem using Dp. 2) Heap Data Structure 3) Heap sort algorithms	Quiz 14

Virtual Lab Links:

Array : https://ds1-iiith.vlabs.ac.in/exp/stacks-queues/recap-arrays.html

Stack : https://ds1-iiith.vlabs.ac.in/exp/stacks-queues/stacks/overview.html

Queue: https://ds1-iiith.vlabs.ac.in/exp/stacks-queues/queues/overview.html

Linked List: https://ds1-iiith.vlabs.ac.in/exp/linked-list/index.html

BST: https://ds1-iiith.vlabs.ac.in/exp/binary-search-trees/index.html

DFS: https://ds1-iiith.vlabs.ac.in/exp/tree-traversal/depth-first-traversal/depth-first-traversal-recursive.html

BFS: https://ds1-iiith.vlabs.ac.in/exp/tree-traversal/breadth-first-traversal/breadth-first-traversal-theory.html

MST: https://ds2-iiith.vlabs.ac.in/exp/min-spanning-trees/index.html

Dijikstra’s: https://ds2-iiith.vlabs.ac.in/exp/dijkstra-algorithm/index.html

Evaluation Policy

Internal	Quiz– 30 Marks	30%	100 %
External	End semester Theory and practical Exam. – 70Marks	70%	

Detailed Evaluation Plan

	Assessment	Weightage %
Internals 30%	**Quiz:** Best 7 out of 14 will be considered for internal evaluation.	**30**
External 70%	**End Semester Examination:** Part 1-MCQ (50 Marks) & Descriptive & Practical problems (20 marks) Maximum Duration: 2 hr	**70**
NOTE: 2 Mandatory Assignments (Non graded)		

Faculty Information

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